A high resolution longitudinal dataset of Austrian news media outlet identities transformations and market structure

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2026-03-14

Abstract

Research on news systems requires data that can represent continuity and change simultaneously. Media outlets persist over time, but they also change titles, merge, split, migrate across sectors, alter their primary distribution area, and become embedded in changing undertaking structures. This paper describes a high-resolution longitudinal dataset for the Austrian political information environment that addresses this problem by treating outlet identity as a formal data object rather than as a stable brand label. The dataset links time-bounded outlet identities to diachronic transformations, synchronic structural relations, media undertakings, annual market indicators, sector-level audience indicators, and structured source metadata. Its central methodological feature is rule-based identity resolution: title changes, amalgamations, new distribution areas, sector changes, and substantial publication interruptions trigger new outlet records instead of being collapsed into a single nominal series. The release further preserves relation typologies, date-precision fields, missing-value semantics, annual validity conventions, and explicit additivity rules for market indicators. Taken together, these layers yield a reusable historical graph of Austrian news media that supports longitudinal reconstruction, concentration analysis, territorial comparison, and linkage to external datasets and registries.

1 Background & Summary

Datasets on news media are often strong on enumeration and weak on temporality. They list currently active brands, current legal entities, or single-year indicators, but they rarely encode the formal transitions by which one outlet succeeds another, two outlets are amalgamated into a new one, a regional title becomes a different territorial unit, or a print publication migrates into another sector. This limitation affects both historical work and contemporary comparative analysis. Where outlet identity is treated as self-evident, downstream research inherits unstable units, opaque continuity assumptions, and misleading aggregation.

The Austrian case makes these problems especially visible. The political information environment has been shaped by long-run discontinuities, sectoral differentiation, territorial segmentation, post-war reorganization, commercialization, and cross-media integration. A dataset that simply counts titles or contemporary organizations cannot capture these transformations. Nor is it sufficient to record only legal entities, because editorially recognizable outlets, market units, and undertaking structures are related but not identical levels of analysis.

The dataset described here was developed within the Austrian News Media Infrastructure project to address that problem for Austria's political information environment. Its conceptual horizon extends from the mid-nineteenth century to the present. The first operational layer concerns media supply: media outlets, their relations, their associated undertakings, and linked market and audience indicators. The design premise is that outlet identity must be made explicit and reproducible. The release therefore models a media outlet as a bounded entity defined jointly by title, period, primary distribution area, and sector, rather than as a name string that passively inherits all historical changes attached to it.

This identity-first design is the principal scientific contribution of the dataset. The release does not simply enumerate outlets; it formalizes the conditions under which a new outlet identity comes into existence. Title change, amalgamation into a new outlet, change in primary distribution area, migration into another sector, and sufficiently long interruption of publication all generate new base units. By contrast, the model also preserves forms of historical linkage that do not collapse distinct units into one, such as split-offs, offshoots, umbrella structures, editorial alliances, and collaboration. The result is a relational dataset in which continuity, succession, co-existence, and hierarchy are analytically distinct rather than conflated.

The public release is therefore structured as a linked set of data objects rather than as a single flat table. It includes outlet identities, diachronic outlet relations, synchronic outlet relations, media undertakings, undertaking-to-outlet relations, undertaking-to-undertaking relations, annual outlet-level market observations, undertaking-level market observations, sector-level audience indicators, source metadata, and lookup tables. This architecture turns the dataset into more than a catalog: it is a reusable historical media graph with a tabular publication layer. In practical terms, it enables users to move between outlet histories, organizational structures, annual market positions, territorial coverage, and source provenance without having to rebuild those links from scratch.

This layered structure is also what makes the dataset attractive for reuse beyond descriptive media history. The same release can support event-history analysis of outlet succession, concentration and ownership research at undertaking level, territorial comparisons of market structure, graph analysis of structural linkage, and linkage to external registries or newspaper corpora. The dataset is therefore designed to be analytically rich at multiple scales rather than optimized for a single canonical use case.

2 Methods

2.1 Scope of the dataset

The dataset covers media relevant to political communication in Austria within present-day borders. Within the broader ANMI framework, this release addresses media supply and its links to selected market and audience indicators. The conceptual focus is not all communication media, but those media services that contribute to the political information environment in the sense of regular provision of news and current-affairs content under editorial responsibility. The core publication objects are therefore outlet identities, structural relations between outlets, media undertakings, undertaking relations, source records, and time-varying observations.

2.2 Observation model and analytical levels

The release is built around three analytical levels. The first level is the media outlet, defined as an editorially and historically bounded media unit. The second level is the media undertaking, defined as an organizational actor that can own, operate, or otherwise structure one or more outlets. The

third level is the observation layer, where annual market or audience values are attached to outlets, undertakings, or sectors. This separation is necessary because the same empirical phenomenon can appear differently depending on analytical level. A publication title may persist as a public brand while the corresponding undertaking changes repeatedly; conversely, one undertaking may control several distinct outlets simultaneously. The dataset preserves these distinctions rather than forcing all evidence into a single table keyed by title.

2.3 Base unit

The base unit is a single media outlet that is distributed regularly and continuously under a specific title, in a specific period, primarily in a specific distribution area, and in a specific media sector. This definition is not merely descriptive; it is operational. A record is created only when those four dimensions jointly define a coherent observational unit. The aim is to minimize retrospective ambiguity when outlets are linked across time or aggregated across market data.

2.4 Operational rules for creating new outlet identities

New outlet identities are generated by explicit transition rules. A new record is created when the title of an outlet changes, when the outlet is amalgamated with others into a new outlet, when the primary distribution area changes, when the outlet moves into another sector, or when publication is interrupted and later resumed beyond the specified tolerance threshold. Short interruptions are not treated as new identities. In the current specification, interruptions of less than one week, or less than one month for weekly newspapers, are ignored.

These rules distinguish identity change from relational linkage. If an outlet succeeds another because of title change, amalgamation, new distribution area, or new sector, the successor receives a new outlet identifier and the two units are linked diachronically. If an outlet is affected by a split-off, generates an offshoot in another area or sector, or enters a relation with an already continuing outlet, the historical linkage is recorded without collapsing the relevant units into a single record.

This distinction matters because several empirically adjacent transformations are analytically non-equivalent. A title change implies continuity under a new nominal identity. An amalgamation implies the disappearance of earlier units into a new outlet. A split-off implies the persistence of one unit alongside the emergence of another. An offshoot captures lateral derivation across territory or sector. The dataset makes these distinctions explicit so that later users do not have to infer them from comments or secondary literature.

2.5 Relation model

The dataset distinguishes diachronic from synchronic outlet relations. Diachronic relations encode temporally ordered transitions and include succession, amalgamation, new distribution area, new sector, interruption, split-off, offshoot, and merger. Synchronic relations encode co-existing structures and include main media outlet, editorial alliance, umbrella, and collaboration. Because synchronic relations often have validity periods that do not coincide exactly with outlet life spans, relation periods are stored independently from outlet start and end fields.

The same logic is extended to media undertakings. Undertakings are modeled separately from outlets because they answer a different analytical question. Outlet identities capture editorially and historically bounded media units; undertaking records capture organizational actors. Undertaking-to-outlet relations capture functions such as legally responsible ownership and other business links. Undertaking-to-undertaking relations capture organizational hierarchy or participation, including sole ownership, shareholder status, limited partnership, general partnership, and related forms. In

Triggering condition	New outlet record	Relation type recorded
Title change	yes	succession
Amalgamation into a new outlet	yes	amalgamation
Change in primary distribution area	yes	new distribution area
Migration into another media sector	yes	new sector
Substantial interruption and later resumption	yes	interruption
Split-off from an existing outlet	yes for the split-off unit	split-off
Offshoot in another area or sector	yes for the offshoot unit	offshoot
Co-existing umbrella or collaboration structure	no	synchronic relation

Table 1: Operational logic used to distinguish new outlet identities from historical or structural relations.

methodological terms, this separation prevents outlet identity from being overdetermined by the legal history of undertakings and prevents ownership analysis from being reduced to title histories.

2.6 Identifiers and code systems

The outlet identifier is a six-digit field. In the conceptual specification, the first digit encodes media type, digits two to five are assigned for structured data entry, and the sixth digit can be used to mark closely related outlet groups. The release also uses identifiers for undertakings, relations, outlet-year observations, and sources. Source identifiers are seven digits long and encode source family, reporting period, and year. This design supports direct record-level provenance tracing for annual indicators.

Controlled vocabularies are used for sectors, mandate types, relation types, date precision, and additivity. Missing data are encoded explicitly rather than left blank. Numeric fields distinguish not applicable from not available values, while text fields use a separate convention for absent information. Historical dates can preserve substituted day or month values through dedicated precision fields.

2.7 Time-varying market and audience observations

The release separates time-constant descriptors from time-varying observations. Outlet-year observations include circulation, unique users, national reach, regional reach, national market share, regional market share, source identifiers, and comments. Undertaking-year observations include market revenue and group revenue. Sector-year observations include total reach and informational use.

The annual observation year is interpreted as the year of validity of the data, not simply the publication year of the cited source. This distinction is necessary because annual reports and audit publications may refer to the previous year. The specification therefore treats the year field as an analytical validity year.

This annualization rule is important for comparability. It makes values align to the period they describe rather than to the bibliographic date of the underlying document. Without that distinction,

Relation family	Relation type	Interpretation
Diachronic	succession	later outlet follows earlier outlet because of title change
Diachronic	amalgamation	earlier outlet is absorbed into a newly formed outlet
Diachronic	new distribution area	later outlet represents territorial redefinition of earlier outlet
Diachronic	new sector	later outlet represents sectoral migration of earlier outlet
Diachronic	interruption	outlet reappears after substantial pause in publication
Diachronic	split-off	new outlet branches from existing outlet while analytical distinction is preserved
Diachronic	offshoot	related outlet is founded in another area or sector
Diachronic	merger	equal or near-equal fusion event represented as a historical relation
Synchronic	main media outlet	hierarchical relation between differentiated unit and parent outlet
Synchronic	editorial alliance	concurrent editorial combination or joint structure
Synchronic	umbrella	umbrella or brand-level aggregation of multiple outlets
Synchronic	collaboration	looser continuing co-operation not captured by the previous types

Table 2: Relation families and analytical meaning in the outlet layer.

time-series analysis would mix publication lag with substantive change.

2.8 Additivity rules

Additivity is explicitly encoded because market indicators cannot always be aggregated safely across outlets. The model distinguishes values that are never additive, always additive, additive only at the level of individual federal states, additive at the level of federal states and Austria as a whole, or additive only at the level of supraregional markets and Austria as a whole. This is a methodological feature, not ancillary metadata: it prevents invalid reconstruction of market totals from overlapping outlet structures such as umbrellas, combinations, or nested regional variants.

In practice, the additivity field acts as a guardrail against double counting. If a publication belongs simultaneously to a regional structure and an umbrella-level market unit, its values may be analytically visible in both places but should enter aggregate calculations only once and only under the permitted territorial scope.

2.9 Source model

The source table serves two functions. First, it documents the provenance of market and audience observations by linking them to annual or period-specific evidence. Second, it makes sources analyzable as metadata in their own right. In the current conceptual specification, source families include official statistics, media audience surveys, web analytics, radio audience data, circulation

Code system	Example values	Function in release
Sector codes	daily newspaper, weekly newspaper, radio, television, online, social media	distinguishes analytical media sectors
Mandate codes	public service, commercial, party, association, official, other	captures outlet mandate or organizational orientation
Date precision	exact date, substituted day, substituted month, substituted year, combined substitutions	preserves historical uncertainty
Additivity rules	not additive, always additive, additive only within specified territorial scopes	constrains valid aggregation
Missing-value rules	not applicable, not available, no textual information provided	distinguishes types of absence

Table 3: Examples of controlled code systems embedded in the release.

audits, and media reports. Because source identifiers are structured, users can reconstruct both the institutional origin and the reporting period of a cited value.

This traceability layer is crucial because the dataset combines values with different evidentiary profiles. Circulation, audience reach, market share, and revenue indicators originate in different institutional ecosystems and are not interchangeable. Structured provenance allows later users to stratify analyses by evidence source instead of treating all numeric values as epistemically equivalent.

2.10 Release architecture

The curation environment preserves nested entity definitions and relation semantics, but the public release is normalized into linked tabular objects. This separation is intentional. It allows the release to remain accessible to repository users while preserving a reproducible transformation path from curated source records to the deposited tables.

The publication layer should therefore be understood as a normalized derivative of a richer curation environment rather than as the primary site of scholarly judgment. This is one reason why comments, status fields, and lookup tables remain important in the released form: they preserve decisions that would otherwise disappear in a minimal flat export.

3 Data Records

The release is organized as a linked set of data objects rather than as a single spreadsheet. Table 4 summarizes the core published objects and the principal variables that define them.

The repository deposit should include UTF-8 CSV files for all tabular objects, a manifest describing release version and file inventory, and codebooks that define variable meanings, coded

Data object	Unit of observation	Primary key	Selected contents
Media outlets	One outlet identity	`id_mo`	Title, sector, mandate, location, primary distribution area, language, start and end fields, date precision, comments
Outlet diachronic relations	One time-ordered relation	`id_relationship`	Predecessor, successor, relation type, event date, comments
Outlet synchronic relations	One concurrent relation	`id_relationship`	Outlet 1, outlet 2, relation type, start and end fields, comments
Media undertakings	One undertaking identity	`id_mu`	Name, foundation year, closing year, comments
Undertaking-outlet relations	One undertaking-to-outlet link	`id_relationship`	Undertaking, outlet, function or relation type, start and end fields, comments
Undertaking-undertaking relations	One undertaking-to-undertaking link	`id_relationship`	Source undertaking, target undertaking, relation type, share, start and end fields, comments
Outlet market observations	One outlet-year observation	composite	Annual circulation, users, reach, market share, additivity flag, source identifiers, comments
Undertaking market observations	One undertaking-year observation	composite	Annual market revenue, group revenue, source identifiers, comments
Sector audience observations	One sector-year observation	composite	Total reach, use as information source, source identifiers, comments
Sources	One source record	`id_source`	Source family, reporting period, year, title and description metadata
Lookup tables	One coded value	code	Sector codes, mandate codes, region codes, date precision, additivity, relation types

Table 4: Core published objects of the ANMI release.

values, and missing-data conventions. A JSON serialization may additionally be supplied for graph-oriented workflows, but the tabular objects should remain canonical for archival publication and reuse. In combination, the tabular and graph-oriented views allow the same release to function as both a high-resolution historical reference dataset and a computational resource for network, market, and longitudinal analysis.

The media outlet table is the key entry point for most reuse, but it should not be treated as self-sufficient. It acquires its analytical meaning only in conjunction with the relation tables, the source table, and the market or audience observation tables. The relation tables are not optional enrichments; they are part of the unit definition. Likewise, the source table is not supplementary metadata. It is required to interpret the observation layer correctly.

The final manuscript should report exact row counts, release version, and temporal coverage for each deposited object. Those values should be generated directly from the deposited release rather than entered manually.

4 Technical Validation

Technical validation addresses whether the release is internally coherent, historically interpretable, and reusable as data. The validation strategy therefore operates on six levels: entity integrity, referential integrity, temporal coherence, domain compliance, source traceability, and aggregation safety.

4.1 Entity integrity

Entity integrity validation checks uniqueness of primary identifiers, absence of duplicate records under the same release version, and conformance to schema-level field requirements. Outlet validation additionally checks the presence of the defining dimensions of identity: title, period, primary distribution area, and sector. Undertaking validation checks identifier uniqueness and consistency of foundation and closing fields. Outlet-year, undertaking-year, and sector-year tables are checked for uniqueness of their composite keys.

At this level, validation also serves as an ontological check. A record that lacks one of the defining outlet dimensions may still be historically interesting, but it should not enter the published release as a fully formed outlet identity without explicit qualification.

4.2 Referential integrity

Every relation record must resolve against published entity tables. Outlet diachronic relations must resolve both source and target outlet identifiers. Outlet synchronic relations must resolve both linked outlet identifiers. Undertaking-to-outlet relations must resolve both undertaking and outlet identifiers. Undertaking-to-undertaking relations must resolve both undertaking identifiers. Market and audience observations must resolve all attached source identifiers against the source table. Hard referential failures should be zero in the deposited release.

Because the dataset is relational by design, referential integrity is not merely a formatting requirement. Broken identifiers degrade the interpretability of continuity chains, undertaking structures, and provenance claims.

4.3 Temporal coherence

Temporal validation checks that start dates do not follow end dates, that ongoing entities follow the documented convention for open-ended validity, and that relation periods do not contradict the life spans of linked entities unless an explicit exception rule is documented. For interruption relations, the interval must bridge the end of the earlier outlet and the beginning of the later outlet according to the specification. For annual observations, the validity year must be interpretable independently from the publication year of the source.

Temporal validation should also identify suspicious boundary patterns, such as apparent successor events without a corresponding relation record, or relation dates that imply impossible overlap or impossible gaps relative to the defining outlet periods.

4.4 Domain compliance

All coded fields are validated against the published lookup tables. This includes sector codes, mandate codes, relation types, date-precision types, and additivity rules. Numerical checks screen for impossible values such as negative circulation, impossible shares, or malformed date values.

Missing-value conventions are also validated to ensure that not applicable and not available are not conflated.

The preservation of separate missing-data states is particularly important for this dataset. Historical absence, current inapplicability, unavailable evidence, and explicit exclusion from aggregation are analytically distinct conditions.

4.5 Traceability and auditability

The release is designed so that every market or audience observation can be traced to a structured source record. Validation therefore checks whether source identifiers are resolvable and whether comments or status fields document cases of estimated, inherited, missing, or non-additive values. This is especially important in historical and market-structure work, where analytically valid reuse depends on knowing whether an absence reflects non-existence, non-observation, or explicit exclusion from aggregation.

4.6 Aggregation safety

Aggregation safety checks whether the additivity rules attached to market observations are logically compatible with the outlet structures in which the observations are embedded. This includes nested umbrellas, combinations, regional differentiation, and overregional units. The purpose of this validation layer is to prevent a formally correct table structure from producing analytically invalid market totals.

5 Data Overview

The scientific value of the release lies less in any single variable than in the way the objects combine. Outlet identities provide the base map of the Austrian news environment. Diachronic relations convert this map into a historical graph of succession and transformation. Synchronic relations capture co-existing structural formations such as umbrella offers and editorial alliances. Undertaking tables connect outlet history to organizational structures, while annual observation tables connect the same entities to circulation, reach, market share, revenue, and sector-level audience exposure.

Taken together, these layers allow at least four broad forms of reuse. First, they support longitudinal reconstruction of the Austrian political information environment, including the explicit modeling of successor chains and interruptions. Second, they support concentration and ownership analysis without collapsing legal actors into outlet titles. Third, they support market analysis that respects territorial and sectoral comparability constraints. Fourth, they support linkage to external datasets that use organizations, brands, years, or source identifiers as their entry points.

The combination of identity resolution, relation modeling, and observation layers also gives the release a degree of analytical granularity that is often absent from media datasets. Instead of one current outlet list plus separate market tables, the release can be queried as a temporally resolved entity network. That makes it suitable not only for static description, but also for event-based analysis, graph extraction, and historically aware feature engineering in downstream computational workflows.

6 Usage Notes

The main analytical choice for reusers concerns the level at which continuity is reconstructed. Outlet identifiers represent bounded outlet identities, not abstract brands. Researchers interested in

long-run continuity should therefore reconstruct successor chains rather than assume that repeated titles define a single uninterrupted unit. Researchers interested in cross-sectional market structure may instead work directly with outlet-year observations. Both approaches are valid, but they answer different questions.

Users should also distinguish brand hierarchy from organizational hierarchy. A title nested in an umbrella structure is not necessarily coextensive with an ownership relation, and a legally responsible undertaking is not necessarily the analytically most relevant unit for studying editorial continuity or market segmentation.

Market indicators should never be aggregated without explicit reference to the additivity field and the accompanying codebook. Some values may be summed only within a restricted territorial scope; others should never be aggregated. Likewise, values coded as unavailable or not applicable should not be recoded to zero.

Finally, the comments fields should not be ignored in advanced use. In a dataset of this kind, comments can carry historically specific explanations of causation, exceptional handling, or boundary cases that cannot always be reduced to closed codes without loss of meaning.

7 Ethics Statement

The dataset is designed around media outlets, organizational actors, source records, and market and audience indicators. The primary release objects therefore concern institutions and publicly observable media structures rather than human-subject responses. Where source materials contain information about natural persons or historically sensitive cases, publication should follow the applicable legal and institutional rules governing archival and public-interest data in Austria and the European Union. The deposited release should document any exclusions, redactions, or access restrictions introduced for that reason.

8 Data Availability

The ANMI dataset described here is intended for deposit in a public repository that provides persistent identifiers, open download access, and long-term preservation. Replace the following placeholder with the final data citation before submission: `[Authors]. [Dataset title]. [Repository]. [DOI or accession] ([year]).` The deposit should contain the published data objects, lookup tables, codebooks, release manifest, and versioned release notes.

9 Code Availability

Code used to transform the curated source records into the published release should be archived alongside the dataset or in a linked public code repository with its own persistent identifier. The final manuscript should cite the exact software release used to generate the deposited data objects.

Funding

Funding details should be inserted here before submission.

Author Contributions

Replace this draft statement with the final authorship declaration before submission. Example: M.A. conceived the dataset design, led the specification of the release model, and drafted the manuscript. [Add co-author contributions here.]

Competing Interests

The author declares no competing interests.